LITERATURE SURVEY

Hybrid GNN-Based NOMA for User Assignment and Slot Pairing

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1. Abstract

Non-Orthogonal Multiple Access (NOMA) allows multiple users to share the same time or frequency resources through superposition coding and successive interference cancellation. Its performance depends on user pairing, power allocation, channel conditions, and resource assignment.

Graph Neural Networks (GNNs) provide a way to represent users and their relationships as a graph and learn resource-management decisions from this structure. This literature survey reviews IEEE Xplore research related to NOMA, user pairing, graph-based radio resource management, and resource allocation.

Based on the reviewed literature, this report identifies a research direction for a hybrid GNN-based approach that combines learned user-pairing decisions with constraint-aware slot assignment. The proposed direction can be evaluated using sum rate, fairness, outage probability, schedule validity, and computational complexity.

Keywords: NOMA, Graph Neural Network, user pairing, slot assignment, resource allocation.

2. Introduction

Wireless communication systems must support increasing data rates and large numbers of connected devices. Non-Orthogonal Multiple Access (NOMA) is a candidate technique for improving resource utilization by allowing multiple users to share a radio resource.

In power-domain NOMA, users may be allocated different power levels. Successive Interference Cancellation (SIC) is used at the receiver to decode selected signals and reduce interference. System performance depends on channel conditions, power allocation, decoding order, and the users sharing a resource.

User pairing is an important part of NOMA scheduling. Selecting compatible users can affect achievable throughput, fairness, and outage performance. Coordinating pairing with the assignment of available time slots can become difficult as the number of users increases.

Graph Neural Networks provide a potential way to learn from relationships between users. In a graph representation, users are nodes and their pairwise channel or interference relationships are edges. A hybrid method can use GNN predictions to guide pairing and then apply a constraint-aware algorithm to assign the selected pairs to slots.

3. Objectives of the Literature Survey

The main objectives are:

- To study NOMA fundamentals and its resource-allocation challenges.
- To understand how user pairing affects NOMA performance.
- To examine GNN-based approaches for wireless resource management.
- To study user selection and resource allocation in multicarrier NOMA.
- To identify a research direction for hybrid GNN-based user assignment and slot pairing.

4. Literature Review

4.1 Paper 1: A Survey on Non-Orthogonal Multiple Access for 5G Networks: Research Challenges and Future Trends

Publication: IEEE Journal on Selected Areas in Communications, 2017.

DOI: 10.1109/JSAC.2017.2725519

IEEE Xplore: View paper on IEEE Xplore

This survey discusses NOMA principles, potential benefits, and research challenges in wireless networks. It provides background on power-domain NOMA, receiver-side interference cancellation, and design issues affecting system performance.

Relevance: This paper provides the theoretical foundation for studying NOMA and motivates the investigation of user selection, power allocation, and resource management.

Research opportunity: The general survey does not itself provide the specific hybrid GNN-based slot scheduler proposed in this project.

4.2 Paper 2: Impact of User Pairing on 5G Nonorthogonal Multiple-Access Downlink Transmissions

Publication: IEEE Transactions on Vehicular Technology, 2016.

DOI: 10.1109/TVT.2015.2480766

IEEE Xplore: View paper on IEEE Xplore

This paper studies the impact of user pairing on downlink NOMA performance. It discusses how pairing strategies and differences in users' channel conditions can influence achievable throughput.

Relevance: It motivates channel-aware user pairing. A GNN could learn pair compatibility from node and edge features derived from user channel conditions.

Research opportunity: A learning-based method can be investigated for generating candidate pairs when the number of possible pairings increases.

4.3 Paper 3: Graph Neural Networks for Scalable Radio Resource Management: Architecture Design and Theoretical Analysis

Publication: IEEE Journal on Selected Areas in Communications, 2021.

DOI: 10.1109/JSAC.2020.3036965

IEEE Xplore: View paper on IEEE Xplore

This work investigates graph neural network architectures for scalable radio resource management. It shows how graph representations can be used to model relationships in wireless systems and learn resource-management decisions.

Relevance: It provides a foundation for representing users as graph nodes and pairwise relationships as edges. The learned representations can be used to estimate candidate pairing or assignment scores.

Research opportunity: The proposed project can adapt graph learning to NOMA user pairing and slot assignment and evaluate its performance against conventional methods.

4.4 Paper 4: User Selection and Resource Allocation Algorithms for Multicarrier NOMA Systems on Downlink Beamforming

Publication: IEEE Access, 2020.

DOI: 10.1109/ACCESS.2020.2982677

IEEE Xplore: View paper on IEEE Xplore

This paper considers user selection and resource allocation in multicarrier NOMA systems. Its focus is relevant to coordinating user grouping with the allocation of radio resources.

Relevance: It supports the need to consider user selection and resource assignment together rather than treating them as entirely independent decisions.

Research opportunity: A GNN can be investigated as a candidate-pair scoring method, with

a separate assignment stage enforcing slot constraints.

4.5 Paper 5: Pairing Distance Resolution and Power Control for Massive Connectivity Improvement in NOMA Systems

Publication: IEEE Transactions on Vehicular Technology, 2020.

DOI: 10.1109/TVT.2020.2975539

IEEE Xplore: View paper on IEEE Xplore

This paper investigates the effects of pairing distance and power control on NOMA connectivity and performance.

Relevance: It motivates using relationships between users to evaluate pairing suitability. Pairwise channel differences and other compatibility features could be considered in the proposed graph representation.

Research opportunity: The project can investigate whether a learned pairing strategy combined with slot assignment provides useful performance and complexity trade-offs.

5. Comparative Analysis

Table 1 compares the five selected papers and their relevance to the proposed project.

No.	Focus	Contribution	Research opportunity
1	NOMA fundamentals	Background and research challenges	Apply learning to a specific scheduling problem
2	User pairing	Shows the importance of pairing decisions	Learn candidate-pair scores using a GNN
3	GNN resource management	Graph-based wireless optimization foundation	Adapt the graph model to NOMA slot scheduling
4	User selection and allocation	Studies resource assignment in multicarrier NOMA	Combine learned pairing with constraint-aware assignment
5	Pairing distance and power control	Motivates channel-aware pair selection	Study joint pairing and slot assignment

Table 1: Comparison of the selected IEEE Xplore papers.

6. Research Gap

The reviewed literature addresses complementary parts of the problem. NOMA studies examine user pairing and resource allocation, while GNN studies provide techniques for learning from graph-structured relationships in wireless systems.

A potential research direction is a hybrid framework that combines:

- A graph representation of users and pairwise channel or interference relationships.
- GNN-based prediction of candidate pairing scores.
- A constraint-aware procedure for assigning pairs to available slots.
- Evaluation using sum rate, fairness, outage probability, schedule validity, and runtime.

This is a potential research gap inferred from the selected papers, not a claim that no previous work has studied similar methods. Further literature searches are needed to establish the novelty of the exact approach.

7. Proposed Methodology

7.1 System Model

Consider a single-cell network with multiple users and a limited number of time slots. The scheduler selects compatible users and assigns them to available slots while respecting the project's resource constraints.

7.2 Graph Construction

Represent the users as a graph

$$G = (V, E),$$

where V is the set of users and E is the set of candidate relationships between users.

Possible node features include channel gain, distance, and target rate. Possible edge features include channel-gain difference, pair compatibility, and an interference-related measure. Feature definitions should be checked to ensure that they provide useful information for the pairing task.

7.3 GNN-Based Pairing

The GNN aggregates information from connected users and learns representations that can be used to estimate candidate pairing scores. The resulting scores guide the pairing stage.

7.4 Slot Assignment

A matching or assignment algorithm selects pairs using the GNN scores and checks the assignment constraints. This stage is important because a high predicted pairing score does not guarantee that the complete schedule is valid.

7.5 Performance Metrics

The approach can be evaluated using the following metrics:

- **Sum rate:** total achievable data rate.
- **Fairness:** for example, Jain's fairness index.
- **Outage probability:** fraction of users below their target rate.
- **Schedule validity:** fraction of schedules satisfying all constraints.
- **Runtime:** time required to make a scheduling decision.

Results should be compared with suitable conventional pairing and resource-allocation baselines using consistent simulation conditions.

8. Conclusion

This literature survey reviewed five IEEE Xplore papers related to NOMA, user pairing, GNN-based radio resource management, and resource allocation. The selected studies provide a foundation for investigating a hybrid GNN-based approach to user assignment and slot pairing.

The proposed direction is to use a GNN to estimate promising user pairs and then apply a constraint-aware assignment algorithm to allocate pairs to slots. The effectiveness of this approach must be established through experiments that measure throughput, fairness, outage, schedule validity, and runtime.

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